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ALL INDIA INTEGRATED TEST SERIES

FIITJEE

# JEE(Advanced)-2016

## ANSWERS, HINTS & SOLUTIONS
## PART TEST-III
## (Paper - 1)

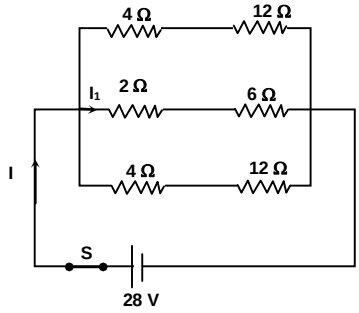
| Q. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | **B, C** | **A, C** | **A, B, C, D** |
| 2. | **B** | **B, C** | **A, C, D** |
| 3. | **A** | **A, B, D** | **B, D** |
| 4. | **B** | **B, C, D** | **B, D** |
| 5. | **A, B** | **A, B, D** | **A, B, D** |
| 6. | **A, C, D** | **A, C** | **C, D** |
| 7. | **A, C** | **B, C, D** | **B, C** |
| 8. | **B, D** | **C, D** | **A, B, C.** |
| 9. | **B** | **A, B** | **A, D** |
| 10. | **A** | **A, D** | **A, B** |
| 1. | **3** | **3** | **3** |
| 2. | **6** | 4 | **3** |
| 3. | **6** | **4** | **4** |
| 4. | **2** | **9** | **7** |
| 5. | **4** | **3** | **2** |
| 6. | **8** | **3** | **2** |
| 7. | **7** | **4** | **6** |
| 8. | **1** | **6** | **0** |
| 9. | **3** | **3** | **1** |
| 10. | **2** | **3** | **5** |

# Physics

## PART – I

## SECTION – A

1. **B, C**
Equivalent circuit is




2. **B**

Loss of energy is $\frac{1}{2}C_eE^2$

3. **A**
$V_A=V_C$
c=a+b

4. **B**

Current through Battery, $I=\frac{V}{R}+\frac{V}{2R}e^{-\frac{t}{2RC}}$.

5. **A, B**

Final common potential across the capacitors is $\frac{2E_oC}{3C}=\frac{2}{3}E_o$ after a long time.

6. **A, C, D**
$\vec{F}_m=q\left(\vec{v}\times\vec{B}\right)$ and apply Newton's second law of motion.

7. **A, C**
As four resistance forms a balanced Wheatstone bridge so potential difference across capacitor $C_1$ is zero.
Charge on $C_1$ = 0
Charge on $C_2$ = $4x10^{-6}$ x 8 = 32 μC
Charge on $C_3$ = $8x10^{-6}$ x 4 = 32 μC
i = 12/6 = 2 A
When $k_2$ is closed.
Potential difference across $C_2$= 4V
So change on it is $4x10^{-6}$ x 4 = 16μC
Potential difference across $C_3$ = 8V
So change on it is $8x10^{-6}x8$ = 64 μC
i = 12/6 = 2A

8. **B, D**
$\vec{F}_m=q(\vec{v}\times\vec{B})$.

9. **B**
$\phi=\vec{E}.\vec{A}$

10. **A**
B = $\mu_o$ni

$n=\frac{B}{\mu_o i}=\frac{1}{4\pi\times10^{-7}\times1}=\frac{1}{4\pi}\cdot10^7$ turns / m ⟶ B = $\mu_o$ni

## SECTION – C

1. **3**

Acceleration and magnetic field are perpendicular to each other thus $\vec{B}.\vec{a} = 0$

2. **6**

Current through 30Ω is $\frac{12}{40}$ and apply loop rule then charge on the capacitor $C_2$ is 60μC.

3. **6**

M = IA

4. **2**

$\int \tau dt = NAB\int idt$

$I\omega_0 = NABQ$ .....(i)

$\frac{1}{2}I\omega_0^2 = \frac{1}{2}C\theta_{max}^2$ .....(ii)

From equation (i) and (ii)

$\theta_{max} = \frac{NABQ}{\sqrt{CI}}$

5. **4**

$B_{net} = \frac{\mu_0 j\sqrt{2}}{2}$

6. **8**

Apply KVL & symmetry charge on the capacitor 1μF is zero.

7. **7**

$B_0 = \frac{\mu_0 I}{4r} + 2 : \left( \frac{\mu_0}{4\pi}.\frac{I}{\frac{r}{\sqrt{2}}}.2\sin 45° \right) = 7.14 \times 10^{-6}T$

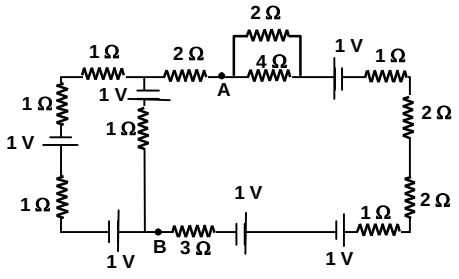
8. **1**

$B\int_0^R \frac{\omega}{2\pi} Q \frac{2\pi r}{\pi R^2} dr\, \pi r^2 = \frac{T_0 D}{2}$

$\Rightarrow \quad \omega = \frac{T_0 D}{QR^2 B}.$

9. **3**

According to Kirchoff's current rule. Current passing through A towards left = current passing through B toward right.



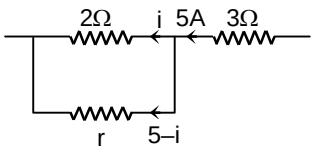

10. **2**

Potential difference across 3Ω resistance is 20V – 5V = 15V

$\Rightarrow$ Current through 3Ω resistance = 5 Ampere.

5 Volt = (5 – i)r = 2i

$\Rightarrow$ r = 2Ω



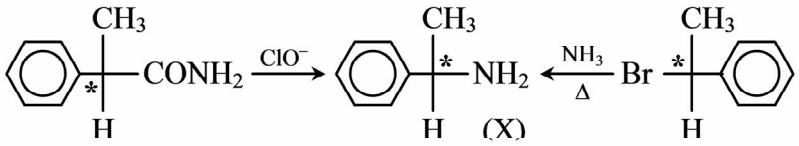
# Chemistry

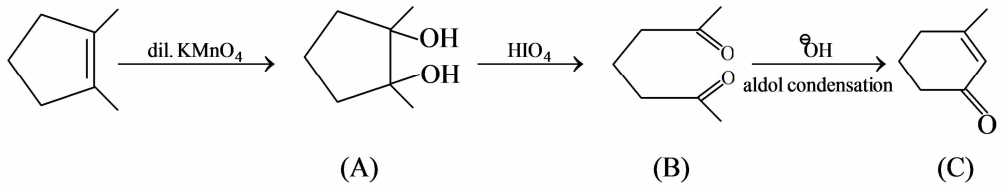
## PART – II

### SECTION – A

4. $NaBH_4$ does not reduce ester.

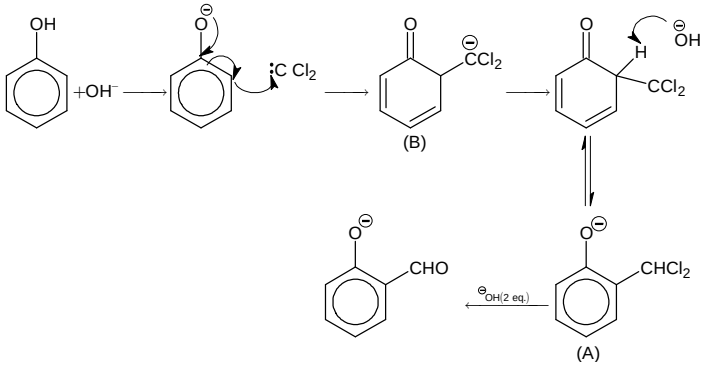
5. Galactose is empimer of D-(+)-glucose. It is monomer.

7. 


8. 


9. $CHCl_3 \xrightarrow{OH^-} CCl_3^- + H_2O$

$CCl_3^- \longrightarrow :CCl_2 + Cl^-$

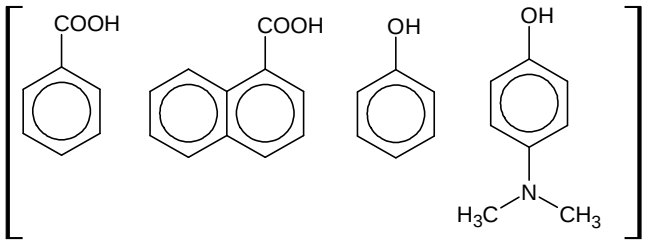



(C) cannot exist in presence of $OH^-$.

### SECTION – C

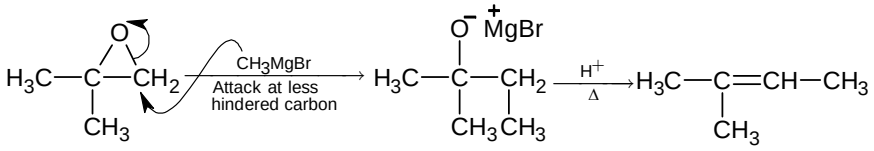
1. One with keto and two with ester group (lactone).

2. All are stereoisomers.

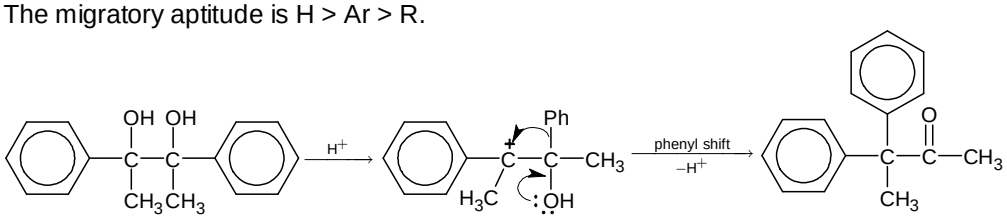
3. 

are soluble in alkalies.

4.




5. Pinacol-Pinacolone rearrangement reaction.
The migratory aptitude is H > Ar > R.



# Mathematics

## PART – III

### SECTION – A

2. $f(x) = (x-1)^4 (x-2)^n$; $n \in N$
$f'(x) = 4(x-1)^3 [x-2)^n + (x-1)^4 n(x-2)^{n-1}$
$= (x-1)^3 (x-2)^{n-1} [(n+4)x - (n+8)]$
If n is odd then $f'(x) > 0$ and if $x < 1$ and sufficiently close to 1
$f'(x) < 0$ if $x > 1$ and sufficiently close to 1
$\therefore$ $x = 1$ is point of local maxima
Similarly if n is even then $x = 1$ is a point of local minima further if n is even then $f'(x) < 0$ for $x < 2$ and sufficiently close to 2 and $f'(x) > 0$ for $x > 2$ and sufficiently close to $2 \Rightarrow x = 2$ local minima.

3. $f(x) = 2x^3 - 3(a+1)x^2 + 6ax - 12$
$f'(x) = 6\{x^2 - (a+1)x + a\} = 0$
$6\{(x-1)(x-a)\} = 0$
$f''(x) = 12x - 6(a+1)$
$f''(1) = 6 - 6a > 0$
$f''(a) = 12a - 6a - 6 = 6a - 6$
if $a < 1$ then $x = a$ is a local max
$2a = 1 \therefore a = \frac{1}{2}$
if $a > 1$ then $x =$ a is a local minima $a = 2$.

5. $$f(x) = \begin{cases} \frac{1}{x+1} & 0 \le x < 1 \\ \frac{2}{x} & 1 \le x < 2 \\ \frac{3}{x-1} & 2 \le x < \frac{5}{2} \end{cases}$$

8. $$f'(x) = \frac{x - \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$$
for horizontal tangent $\ln x = 1 \Rightarrow x = 1$
if cut the x axis
$$\frac{\ln x}{x} = 0 \Rightarrow x = 1$$
$f'(x)$ is +ve $x \in (0, e)$ and $f'(x)$ is –ve if $x \in (0, \infty)$
$f(x)$ is not monotonic
for vertical tangent $f'(x) = \infty$
$\Rightarrow \quad \frac{1-\ln x}{x^2} = \infty \Rightarrow \frac{x^2}{1-\ln x} = 0$
$\Rightarrow$ $x = 0$ which is not in domain.

9. $F'(x) = \frac{-2}{x^3}\int_0^x \{4t^2 - 2F(t)\}dt + \frac{1}{x^2}\left(4x^2 - 2F'(x)\right)$ put $x = 4$
$$F'(4) = 0 + \frac{1}{16}\left[4(16) - 2F'(4)\right]$$

$F'(4) = \frac{32}{9}$

for option D

$f(x) = \frac{1}{x^2}\int_0^k \left[4t^2 - F'(t)\right]dt = \frac{1}{x^2}\left[\frac{4t^3}{3} - 2F'(t)\right]_4^x$

$F(8) = \frac{1}{8^2}\left[\frac{4.8^3}{3} - 2F(8) - \frac{4.4^3}{3} + 2F(4)\right]$

$F(8) = \frac{32}{33}\left[\frac{28}{3}\right]$.

10. (A) $\lim\limits_{x\to 0^-}\left|x+\frac{1}{2}\right| = 0 \quad \lim\limits_{x\to 0^+}\left|2x+\frac{1}{3}\right| = 0 \quad \therefore \lim\limits_{x\to 0}\left[f(x)\right] = 0$

(B) $\lim\limits_{x\to 0^-}\left(x+\frac{1}{2}\right) = \frac{1}{2}; \lim\limits_{x\to 0^+}\left(2x+\frac{1}{3}\right) = \frac{1}{3} \quad \therefore f(0) \neq f(0^+)$

(C) $\lim\limits_{x\to 0} f(x)$ does not exist so $\left|\lim\limits_{x\to 0} f(x)\right|$ no exist

(D) $\lim\limits_{x\to 0^-}\frac{\left|x+\frac{1}{2}\right|}{x} = 0 ; \lim\limits_{x\to 0^+}\frac{\left|2x+\frac{1}{3}\right|}{x} = 0$

$\therefore \lim\limits_{x\to 0}\frac{\left[f(x)\right]}{x}$.

## SECTION – C

1. $x = \cos\theta \; y = \sin\theta$

$x + y = \sin\theta + \cos\theta$

2. $f\left(x+\frac{1}{2}\right) + f\left(x-\frac{1}{2}\right) = f(x)$

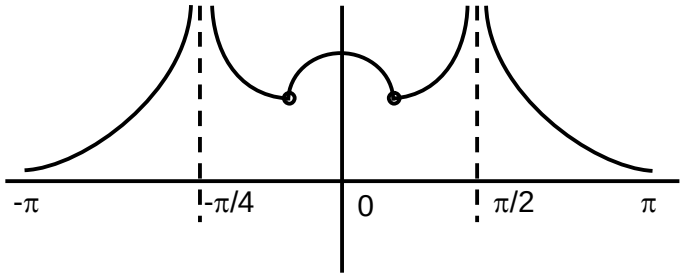
$f(x+1) + f(x) = f\left(x+\frac{1}{x}\right)$

$\Rightarrow \quad f(x+1) + f\left(x-\frac{1}{2}\right) = 0$

$f\left(x+\frac{3}{2}\right) = -f(x)$

$\Rightarrow \quad f(x+3) = -f\left(x+\frac{3}{2}\right) = f(x)$

3. Function is not differentiable in $(-\pi, \pi)$ at points $\left\{-\frac{\pi}{4}, \frac{\pi}{4}\right\}$ and $\left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$.




4. Point of intersection are $\left(\pm\sqrt{2}, 1\right)$

$y = |x^2 - 1|$ and $y = |x^2 - 3|$

$\frac{dy}{dx} = 2x$ $\frac{dy}{dx} = -2x$

$m_1 = \left.\frac{dy}{dx}\right|_{(\sqrt{2},1)} = 2\sqrt{2}$ and $m_2 = \left.\frac{dy}{dx}\right|_{(\sqrt{2},1)} = -2\sqrt{2}$

$\tan\theta = \left|\frac{4\sqrt{2}}{7}\right|$.

5. $I = \int \frac{\cos^4 x\, dx}{\sin^3 x\left(\sin^5 x + \cos^5 x\right)^{3/5}} = \int \frac{\cos^4 x\, dx}{\sin^6 x\left(1 + \cot^5 x\right)^{3/5}}$

$= -\frac{1}{5}\int \frac{dt}{t^{3/5}} = -\frac{1}{2}\left(1 + \cot^5 x\right)^{\frac{2}{5}} + c$.

6. $\int_{-\pi}^{\pi}\left(\cos px - \sin qx\right)^2 dx = \int_0^{\pi}\left(\cos^2 px + \sin^2 qx - 2\sin qx\cos px\right)dx$

$= 2\int_0^{\pi}\left(\frac{1+\cos 2px}{2} + \frac{1-\cos 2qx}{2}\right)dx$

7. $\Rightarrow \lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{\sqrt{4n^2 - r^2}} = \lim_{n\to\infty}\sum_{r=0}^{n-1}\frac{1}{n}\cdot\frac{1}{\sqrt{4 - \left(\frac{r}{n}\right)^2}}$

$= \int_0^1 \frac{dx}{\sqrt{4 - x^2}} = \left[\sin^{-1}\left(\frac{x}{2}\right)\right]_0^1 = \frac{\pi}{6} - 0 = \frac{\pi}{6}$.

9. Volume $= \frac{1}{3}$ (area of base x height)

$= \frac{1}{3}\left(\frac{\sqrt{3}}{4}a^2\right) \times 6 \times h$

$a = \sqrt{1 - h^2}$

$v = \frac{\sqrt{3}}{2}h\left(1 - h^2\right)$

$v' = \frac{\sqrt{3}}{2}(1 - h^2 + h(-2h)] = 0$

$h = \frac{1}{\sqrt{3}}$

$v'' < 0$ at $h = \frac{1}{\sqrt{3}}$

$h = \frac{1}{\sqrt{3}}$.

10. $F'(x) = \left[ f\left(\frac{x}{2}\right) f'\left(\frac{x}{2}\right) + g\left(\frac{x}{2}\right) g'\left(\frac{x}{2}\right) \right]$

here $g(x) = f'(x)$

$F'(x) = f\left(\frac{x}{2}\right) g\left(\frac{x}{2}\right) - f\left(\frac{x}{2}\right) g\left(\frac{x}{2}\right) = 0$

$\Rightarrow$ F(x) is a constant function.